

Solution Requirements

Functional and non-functional requirements

Date	30 September 2026
Team ID	SWDIT-2026-9380
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Team	Lavinraj (Team Leader), Subash, Rahul, Hariharansudhan
Maximum Marks	4 Marks

Functional requirements

ID	Requirement	Description	Priority
FR-01	Profile capture	Collect goal, experience, weekly availability, session length, equipment and preferences.	High
FR-02	Input validation	Reject incomplete, contradictory or unsafe values with clear messages.	High
FR-03	AI plan generation	Use a Gemini model to create a structured weekly plan.	High
FR-04	Plan structure	Show overview, daily exercises, sets/ reps or duration, rest, warm-up and cool-down.	High
FR-05	Adaptation	Offer easier alternatives and equipment substitutions.	High
FR-06	Regeneration	Regenerate after a preference or schedule change.	Medium
FR-07	Progress check-in	Capture difficulty and completion to guide the next plan.	Medium
FR-08	Export	Produce a printable/shareable plan without exposing secrets.	Medium
FR-09	Safety boundary	Display scope limits and recommend qualified help for pain, injury, pregnancy or medical conditions.	High

Non-functional requirements

ID	Category	Target / acceptance criterion
NFR-01	Performance	Interface actions respond within 200 ms; 95% of AI plan requests complete within 12 seconds under expected load.
NFR-02	Reliability	Graceful timeout and retry message; malformed model output never appears as a valid plan.
NFR-03	Security	API key remains server-side; inputs are validated; output is escaped; logs exclude sensitive profile text.
NFR-04	Privacy	Collect minimum data, explain purpose, and allow plan-history deletion when persistence is enabled.
NFR-05	Accessibility	Keyboard-operable controls, visible focus, semantic headings, labels and WCAG AA contrast.
NFR-06	Maintainability	Separated UI, validation, prompt, model adapter and plan-formatting modules with tests.
NFR-07	Compatibility	Usable on current mobile and desktop browsers from 360 px width upward.